O que é um Átomo?

Nuclear Explicado

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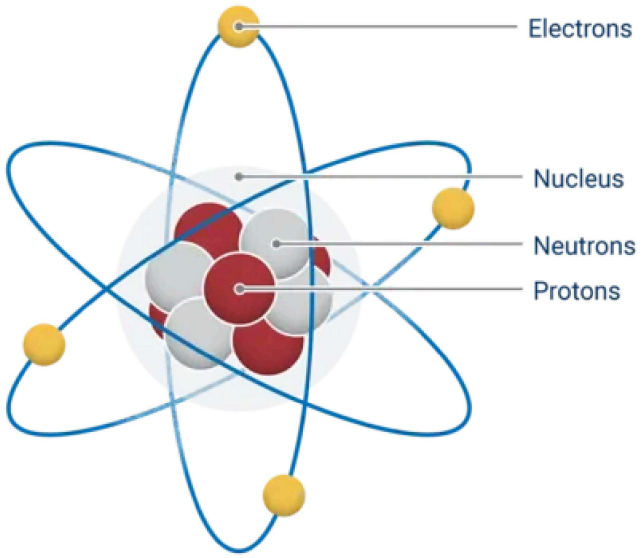
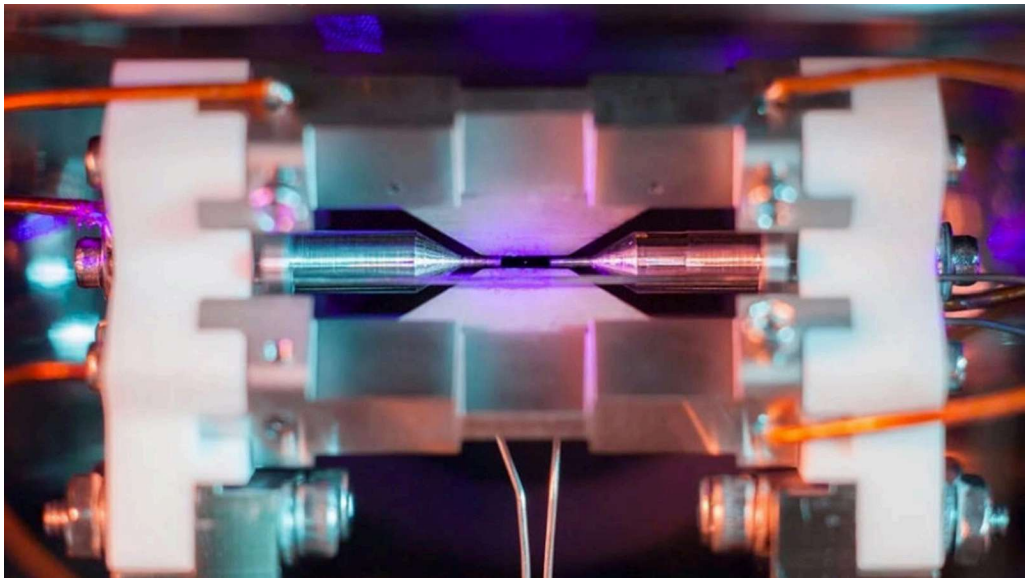


Um átomo é a menor unidade de um elemento que retém suas propriedades químicas. É feito de prótons, nêutrons e elétrons. (Imagem: M. Magnaye)

Os átomos são os blocos de construção da matéria. Tudo ao nosso redor — do ar e da água às rochas, plantas e animais — assim como tudo dentro do nosso corpo, é feito de átomos.

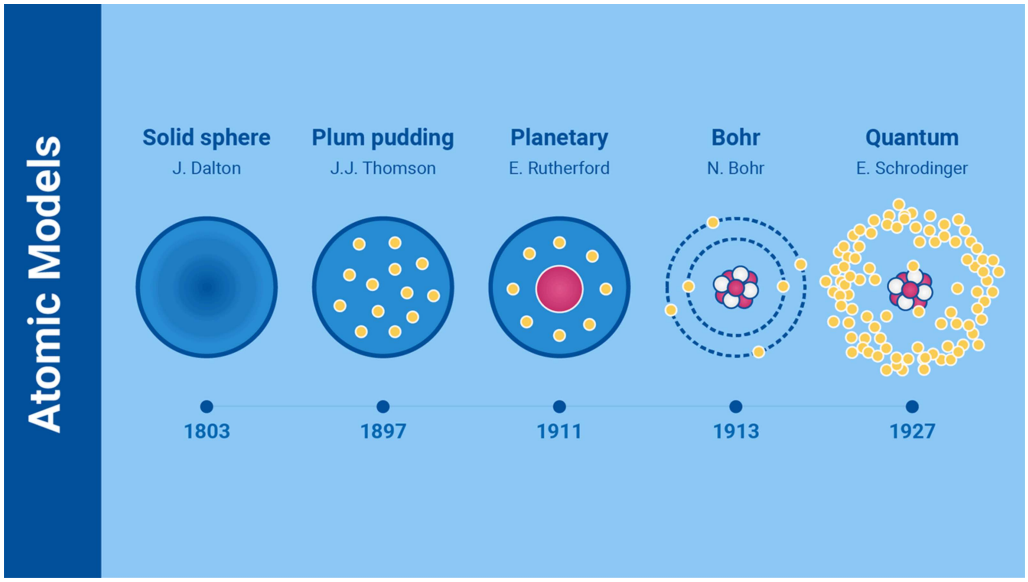
Elas são muito pequenas, as menores unidades de um elemento que retêm as propriedades químicas do elemento. Os gregos antigos acreditavam que eram as menores partículas

existentes, e a palavra ‘átomo’ é derivada de ‘indivisível’ em grego. Um único fio de cabelo humano tem a espessura de 500.000 átomos de carbono empilhados uns sobre os outros.



Este único átomo do estrôncio metálico é visível nesta fotografia porque absorveu e reemitiu a luz de um laser. Os eletrodos na imagem estão separados por dois milímetros. (Foto: David Nadlinger/Universidade de Oxford)

Os átomos não podem ser vistos a olho nu, ou mesmo sob um microscópio padrão. Um átomo é pequeno demais para desviar ondas de luz visíveis, o que significa que ele não aparecerá em microscópios com foco de luz. Os átomos podem ser visualizados sob um microscópio eletrônico, que gera ondas de elétrons que podem interagir com os átomos. Na imagem acima, o átomo é ‘visível’ porque absorveu e reemitiu a luz de um laser.



Como são os átomos? Os cientistas mudaram de ideia sobre o séculos. (Infográfico: M. Magnaye)

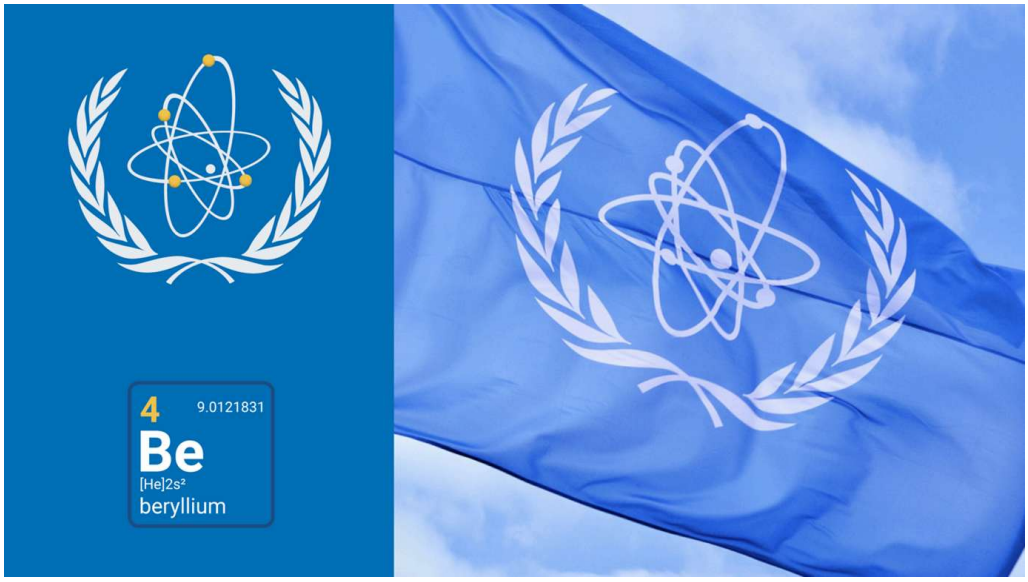
Do que são feitos os átomos?

Each atom consists of three types of particles: protons, neutrons and electrons. At the centre of an atom is a dense nucleus, which contains protons and neutrons, and is much smaller than the entire atom. If the nucleus of the atom were the size of a marble, the atom would be the size of a sports stadium.

Protons have a positive electrical charge, while neutrons are neutral. The nucleus stays together due to the ‘nuclear force’. This force binds the protons and neutrons together at distances close to the size of the nucleus. The nuclear force at this distance is much stronger than the electrical repulsion between the protons (as they have equal charges, they would otherwise repel each other). At larger distances this nuclear force rapidly becomes insignificantly small.

The number of protons in an atom's nucleus determines which element it is. For example, an atom with one proton is hydrogen, while an atom with eight protons is oxygen.

Surrounding the nucleus is a cloud of electrons — negatively charged particles. The atomic nucleus and the electrons are bound together by Coulomb force interactions – the forces in physics that describe the repulsion or attraction between these charged particles. However, when an electron gains energy, it can separate from the atom, causing the atom to become a positively charged ion.

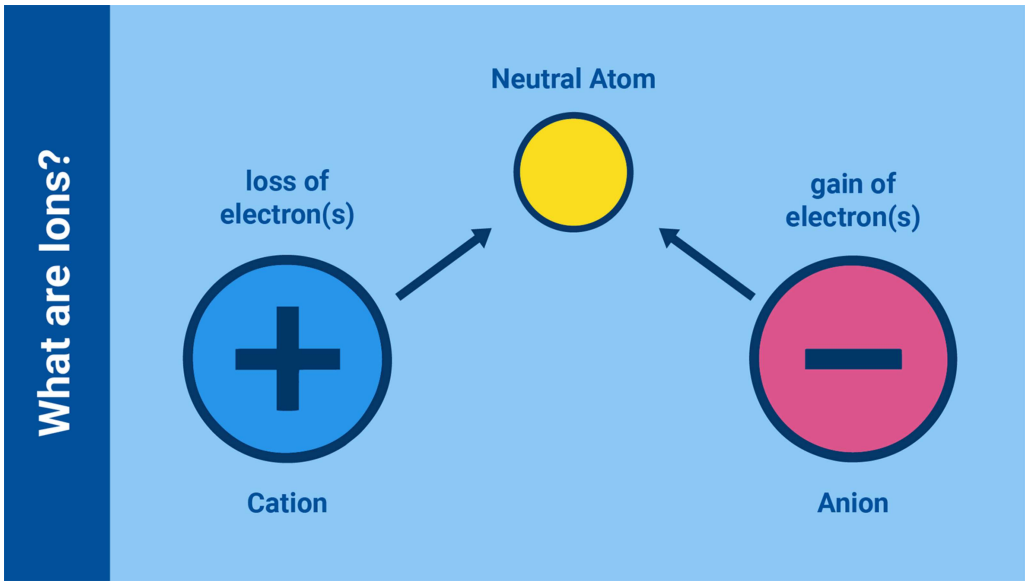


The atom at the centre of the IAEA's logo has four electrons – meaning it is Beryllium if it is neutral and not ionized. (Infographic: M. Magnaye)

What are Ions?

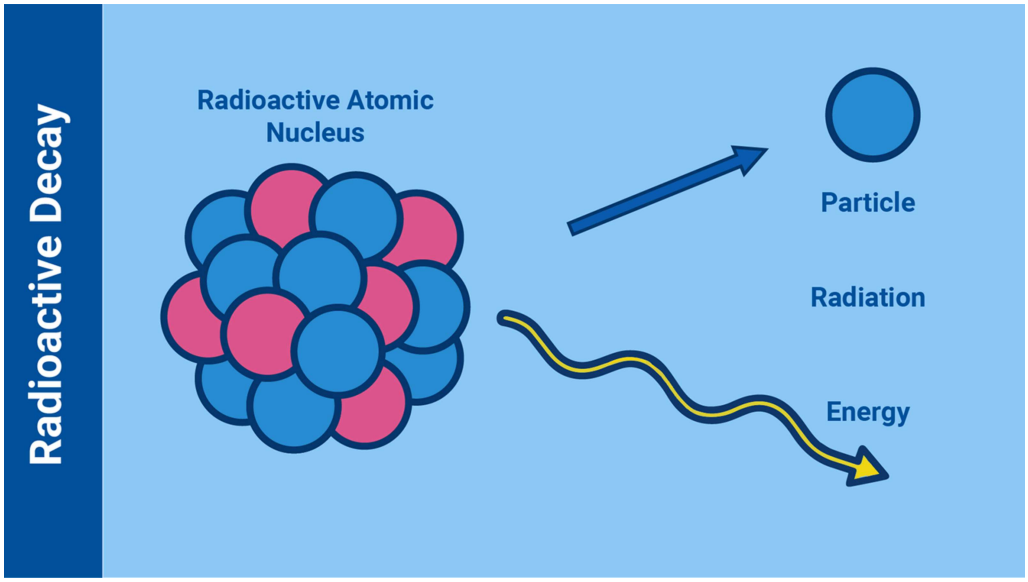
Atoms with the same number of negatively charged electrons and positively charged protons are neutral, as the charges

cancel each other out. If an atom gains or loses electrons it becomes an ion.



(Infographic: M. Magnaye)

While the electric field of a neutral atom is weak, an electrically charged or ionized atom has a strong electrical field, making it strongly attracted to oppositely-charged ions and molecules. Atoms can be ionized by collisions with other atoms, ions and subatomic particles. They can also be ionized by exposure to gamma or X ray radiation. Ionizing radiation refers to radiation that has enough energy to break an electron away from an atom. It can also chemically alter material, for example damaging DNA in living tissue.

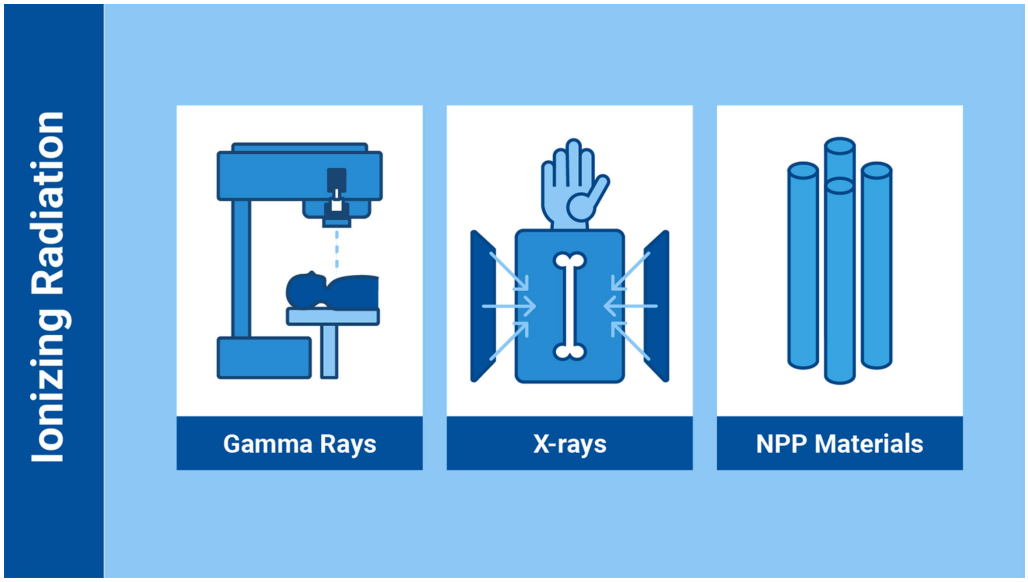


(Infographic: M. Magnaye)

Most atoms on Earth are stable, mainly thanks to a balanced composition of particles (neutrons and protons) in their nucleus.

However, in some types of unstable atoms, the composition of the number of protons and neutrons in their nucleus does not allow them to hold those particles together. In this case, the atom 'decays', and releases energy in the form of radiation (for example alpha particles, beta particles, gamma rays or neutrons), which, when safely harnessed and used, can produce various benefits.

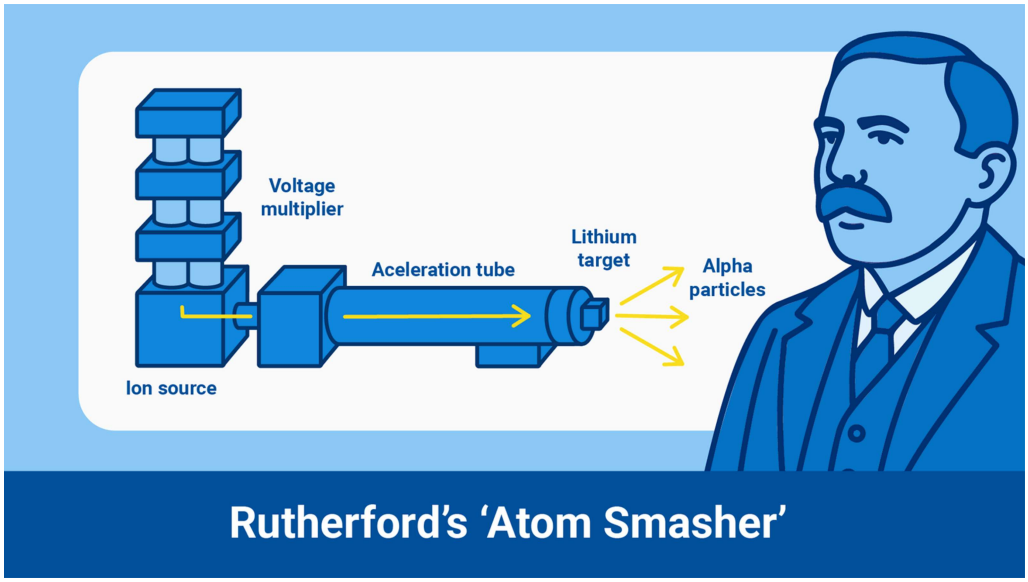
Read more: What are Isotopes?



(Infographic: M. Magnaye)

Ernest Rutherford: Inventor of the ‘Atom Smasher’

In 1917, a scientist called Ernest Rutherford discovered that by blasting beams of radioactive alpha particles into nitrogen gas, the nitrogen atom could be transmuted into oxygen while ejecting a hydrogen nucleus. This subatomic particle (the hydrogen nucleus) was later renamed the proton.



(Infographic: M. Magnaye)

Rutherford's discovery led to the development of the first particle accelerator, initially referred to as an 'atom smasher'. This powerful machine could accelerate charged particles using an electrical field to high energies along a path and used strong magnets to create beams of single charged particles. When the fast-moving particles hit the target (they could go almost as fast as the speed of light), the atoms in the target split apart.

Read more: What are particle accelerators?

Particle accelerators also can be used to create radioactive material by shooting charged particles at atoms to change them into different, unstable atoms, such as technetium-99m for medical imaging and radioisotopes for targeted cancer therapy.

Read more about radioisotopes here.

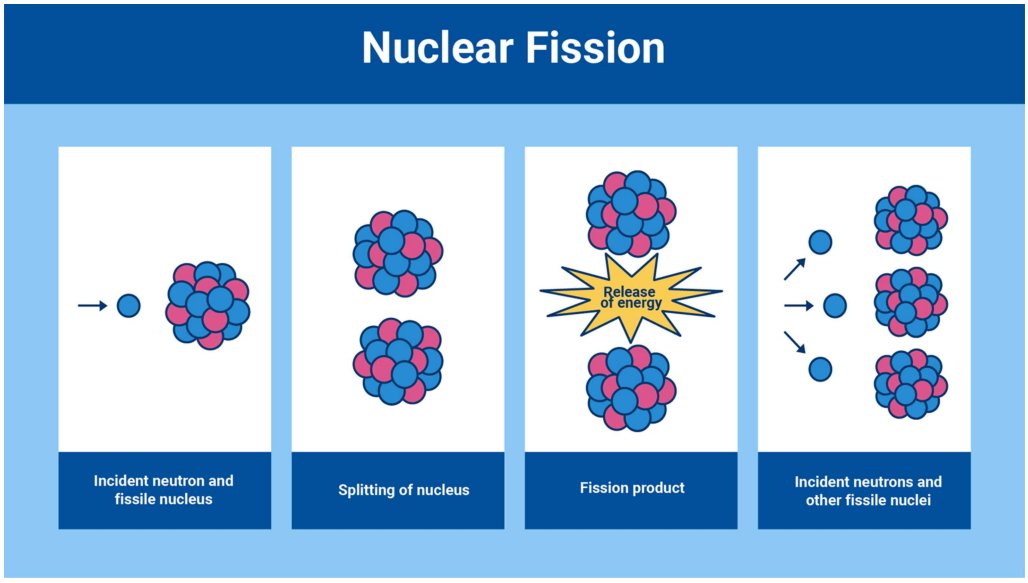
Today, particle accelerators are also used to sterilize medical equipment, research the origins of the universe (for example, at the Large Hadron Collider), as well as to analyse air samples and to enhance materials and make them more resistant to damage. Different types of particle accelerators include ion implanters, electron beam accelerators, cyclotrons, synchrotrons, linear accelerators (Linacs) and electrostatic accelerators.

Splitting the Atom: Nuclear Fission

In the 1930s, scientists found out that if a neutron is fired into certain uranium atoms, they could split into two and emit a certain number of neutrons, releasing a huge amount of energy along the way. This is called fission, from the Latin word for 'split'.

Uranium, with 92 protons, has the highest atomic number of all naturally occurring elements on Earth. Uranium-235 is easier to split (fission) than other isotopes because its nucleus is relatively unstable, and readily absorbs a neutron, causing it to break apart into two lighter atoms. However, only 0.7 per cent of uranium found on earth is this type of uranium, described as fissile.

Read more about uranium here.



(Infographic: M. Magnaye)

Fission can be used to create a nuclear chain reaction. Every time a uranium-235 atom is split it releases on average 2.5 neutrons. These can go on to split further fissile nuclei, releasing yet more neutrons. However, these 'fast' neutrons

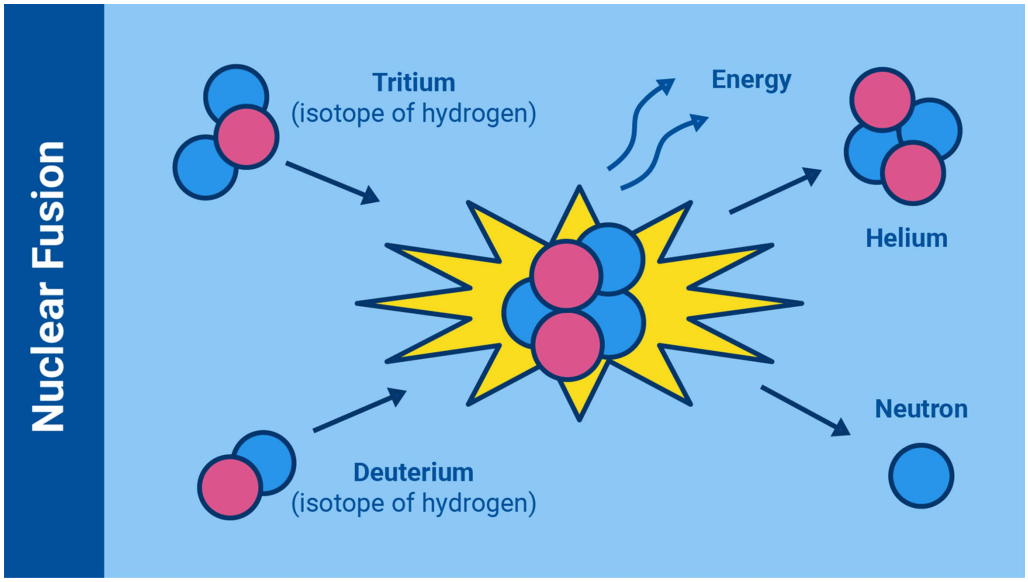
initially travel with too much energy to be effective at causing fission. Using a ‘moderator’ such as water or graphite slows down the neutrons. The neutrons lose most of their energy in collisions with the hydrogen or carbon atoms to become ‘thermal’ or ‘slow’ neutrons which have a much better chance of splitting other uranium nuclei.

The nuclear fission technique is now used to make 10% of the world’s carbon-free energy — as nuclear fission produces no carbon dioxide.

What happens to Atoms in Nuclear Fusion?

Nuclear fusion is the process by which two light atomic nuclei combine to form a single heavier one while releasing massive amounts of energy, a theory first understood in the 1920s.

Fusion reactions take place in a state of matter called plasma — a hot, charged gas made of positive ions and free-moving electrons with unique properties distinct from solids, liquids or gases.



(Infographic: M. Magnaye)

The sun, along with all other stars, is powered by this reaction. To fuse, nuclei need to collide with each other at extremely high temperatures, around one hundred million degrees Celsius. The high temperature provides them with enough energy to overcome their mutual electrical repulsion. Once the nuclei come within a very close range of each other, the attractive nuclear force between them will outweigh the electrical repulsion and allow them to fuse. For this to happen, the nuclei must be confined within a small space to increase the chances of collision. In the sun, the extreme pressure produced by its immense gravity creates the conditions for fusion.

Read more about fusion energy in our explainer ⇢

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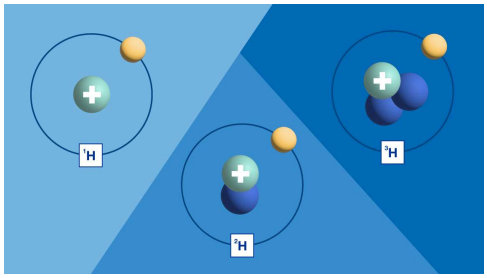

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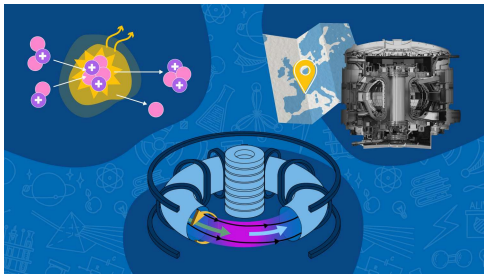

What is Nuclear Fusion?

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Normas e guias

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